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Grow Your Artificial Intelligence Competence

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Artificial intelligence (AI) deployment is growing faster than AI competences. Having worked on AI and competence management across industries, in this article, we provide guidance on growing your AI skills, both hard and soft.

Artificial intelligence (AI) has been around for almost 70 years, with a variety of products and applications over the years. But only recently AI risks are addressed. The European Union's AI law formally took effect on 1 August 2024, covering all AI-driven products and services. Four risk levels are distinguished and drive restrictions. The provisions will come into force over the next two years and demand new competences not only in technology but also ethics and judgement. The term *AI* is a constantly evolving narrative driven by current application scenarios. Historically, it concerned expert systems, while today, it includes autonomous systems and GenAI.[1,2] The quality of AI heavily depends on the competences of those who design, train, and integrate such systems.

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NEED FOR AI COMPETENCE

AI competence needs evolve in parallel to AI technology. For instance, we all know that the quality of the data used for AI learning and customized use matters, but we still have no means to judge data quality and, thus, AI validity and transparency. As there was no way to transparently test or even prove the validity of the generated results, researchers were hesitant to bring this technology to the mass market and rather naive data citizens.[2,3] With the wide public release of rather immature tools, such as ChatGPT and Bard, the AI arms race began at an unprecedented pace—making society

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FROM THE EDITOR

AI improves productivity, but it also creates new risks, as it is neither transparent, deterministic, nor explainable. It is changing what we need to learn and how we need to learn. AI demands fully new hard and soft competences, such as co-piloting. This article introduces critical AI competences and how to grow them. It provides an industry-proven AI competence grid to evolve own AI competences. —*Christof Ebert*

blind to risks. We are now in a catch-up mode to grow AI competences before losing control of AI products. Note that we use the word "competence" in its initial meaning derived from the Latin word "competere," which means "to be able to do something." Competence means the combination of knowledge, skills, abilities and abilities to perform tasks, even if they are situationally new or require innovative action. AI competence thus is the set of skills and capabilities of a person to handle the new and often challenging tasks related to using AI.

AI will not replace humans, we are reassured by politicians. Yet, many studies show that AI will replace those who do not know how to use it.[3,4] If you don't move with time, you will be moved with over time. This holds for individuals as well as entire companies. With GenAI, one-third of the hours worked in Europe and North America today could be automated within the next few years if the current AI adoption rate holds.[4]

Industry predictably reacts to such threats. Over the past months, practically all product releases and announcements have been labeled as "AI inside." Good for the AI supply chain leaders, such as Alphabet, Meta, and Microsoft, with their front ends; Nvidia, with AI GPUs; and companies, such as ASML, with semiconductor production equipment. In the search for business models, many just jump on the bandwagon in a hurry and believe they can undo all past mistakes by expensively licensing GenAI tools.

The Internet has created a digital paradox that is further amplified by AI. With better access to the world's available knowledge, we face increasing difficulties to benefit from and use this knowledge.[5,6] To create and disseminate knowledge is becoming easier with each new technology, while understanding is becoming more and more difficult. Jurgen Habermas had already reflected on the outcomes of this paradox 40 years ago,[7] pointing to the risk to democracy and loss of shared authorities and mutual trust in society, ultimately evolving into different shades of anarchy.

Today, we are amid an arms race across companies and societies. Companies and governments, especially in totalitarian countries without an established balance of powers, have created business models based on fake news, manipulation, and surveillance.[8,9] If society finds business models based on fake news and nonprivacy offensive, democratic systems should react with laws and regulations faster than public opinion is eroded against democratic systems. Although people in democratic countries have the power to vote for politicians who would legally ban such malicious behaviors, they support and multiply such behaviors during the week and on Sunday, demand ethical guidelines.

AI needs an appropriate competence set that goes beyond online videos and GenAI experiments. Such competence growth must start with primary school and evolve toward tailored programs for professionals. The more machine learning systems supposedly dominate thinking and the more human abilities they simulate, the more vigilant and critical people must become.[5] With all the enthusiasm, we need to learn hard competences, such as how to develop, verify, test, and use AI tools. It is even more relevant to grow our software skills, such as making sensible decisions and not blindly trusting technology.[3,6]

GenAI supposedly provides fast content generation for almost any purpose. With the lack of identification and verification of AI work products, fake information and intellectual property violations are fast growing. Experts still easily detect generated outputs with the stereotypical formulations and absurd mistakes. For instance, highly hyped medical AI, such as IBM's Watson and Google's Med-PaLM, look impressive but only with standard questions. Asking a follow-up on sources and accompanying models immediately shows the systems' severe limits. Trusting such platforms will not only reduce our own human expertise but also further squeeze our health-care system.

Lack of competence is the biggest challenge identified in our annual industry survey, conducted with the IEEE Computer Society. At year-end 2023, Vector Consulting contacted industry experts about their challenges and resolutions. With a response rate of 6%, from almost 4,000 recipients from different industry domains, the survey well represents different business models and regions in the world (see "Survey of Industry Trends").

Figure 1 shows the perception of upcoming challenges across industries and regions. The horizontal axis shows short-term challenges, while the vertical axis shows midterm obstacles. Each respondent was allowed to

mention up to five challenges in both dimensions. The sum is thus more than 100%. Innovation and competitiveness are directly impacted. Competence matters to stay competitive in our fast-changing technology landscape. However, many technology companies increasingly suffer from inadequate competences, both hard and soft competences.

SURVEY OF INDUSTRY TRENDS

What's next? Your insights matter, and we invite you to take our short survey on 2025 industry trends. Please give us 2 min of your time and answer few brief questions. After the survey, you will receive exclusive results analysis, enriched with experiences and hands-on advice from current projects. With some luck, you will also receive a free copy of the IEEE book *Global Software and IT*. We will not use your personal data any further. The link to the industry survey is www.vector.com/trends-survey.

FIGURE 1. The lack of competences is the biggest industry challenge.

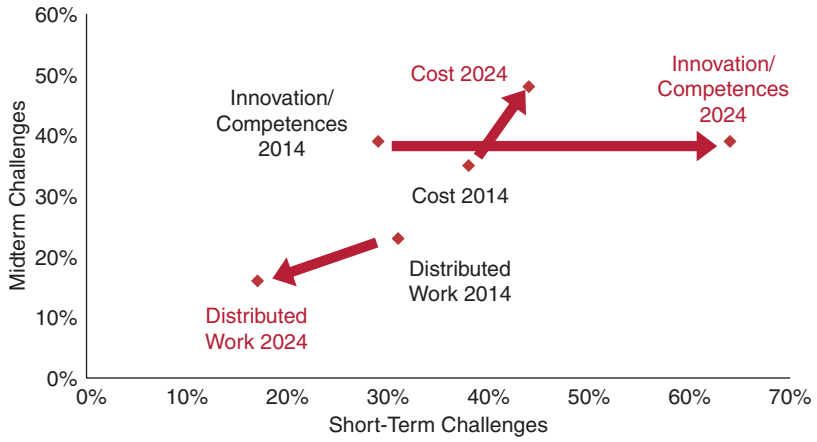

FIGURE 2. A timeline evolution of the competence gap.

INDUSTRY COMPETENCE PROGRAMS

Having worked on many industry competence programs, we started some years ago to systematically grow AI competences, both hard and (!) soft. Note that in this context, when speaking about competence, we mean many dimensions, from technical know-how to business know-why and social know-them.[3,5] Engineers with the right AI competence mix are in demand across domains, from IT to the Internet of Things (IoT), from automotive to finance, medical, and robotics. There is even a legal obligation to grow professional competences because each major standard demands learning and proof of learning.[2] With fast-changing technology, individual soft competences grow in relevance, such as learning to learn, communication skills, culture empathy, and awareness of social and ethical implications of work.

The relevance of having the right competences has been steadily growing over the past 10 years. Figure 2 provides three timelines from our annual surveys over the past decade. With global competition, cost remains a major strategic challenge. Distributed work, on the other hand, is increasingly mastered by companies. The biggest surprise, however, is the fast-growing challenge of having the right competences. It is the biggest short-term challenge across industries today.

As AI competence needs are rapidly evolving, we distinguish and classify levels of learning and methods of interaction. Different levels of learning are distinguished[3,5]:

- *Novice*: observer role and ignorant of rules and dependencies
- *Advanced beginner*: learning facts and analogies
- *Competent*: learning in the sense of understanding
- *Proficient*: learning to further grow
- *Expert*: learning about oneself in the sense of identity learning.

AI is changing what we need to learn and how we need to learn and use this

learning.[1] Mere fact-based learning is replaced in the time of GenAI by understanding which "facts" to trust. Some traditional software tasks will increasingly be enhanced or even replaced by AI.[3,4,5] For instance, software verification and validation will use AI to identify critical feature correlations and corner cases. Automatic pattern matching will be used to detect defects and deviations from requirements, rather than traditional coverage criteria.

Knowledge on an expert level contributes to a person's identity. In mixed teams, it is wise to exchange on such levels of learning distinguishing knowledge at an expert level, superficial knowledge, and nonknowledge or ignorance. Every project has a defined knowledge profile concerning deep and orientation knowledge. If teams share their own respective levels of expertise, they can do a matching of existing and lacking skill levels for their common projects. Being aware of blind spots is a prerequisite of curing them. Such digital ignorance competence is a basis for individual competence evolution programs.[5]

Industry competence programs must individually address implicit and explicit competences. Learning by doing (that is, hands-on formats), though helpful, must be adjusted to needs. A team could report on its respective learning interests and successes, information offers, and further training. Within project reviews, Kanban and Scrum meetings, and team exchanges, you can talk briefly at the beginning about development and learning. A short round on the question What did I learn last? or What did I acquire knowledge about last? stimulates learning by sharing. Sharing a success or failure helps in addressing more implicit competences.

CRITICAL AI COMPETENCES

AI needs critical competences that go way beyond typical engineering and computer science. For instance, identifying the right training data needs profound statistical skills but also soft skills, such as communication, empathy, and negotiation. And it needs legal skills to understand intellectual property and how to avoid bias. Building an AI system needs ethical competences, not just on fairness and judgment but also to identify risks and communicate them—against all odds.

AI competences span a wide range of hard and soft skills and explicit and implicit knowledge. Explicit competences are built with intentional learning of information, such as learning words in a new language. Implicit competences grow in the absence of conscious awareness over time and different episodes. An example is learning how to speak a new language.

AI needs completely new competences, such as copiloting. It is the proficiency to observe a system in its behavior and outputs over time to diagnose malfunctions. It includes vigilance and monitoring, with the sudden switch from observing to alert to action. Often, we easily speak about AI being our copilot. However, we know from autonomous driving that accepting such a copilot requires lots of practice to handle and supervise the copilot. Humans enjoy doing, and if not required, they switch to idle or get into other tasks, as we see with the many accidents in assisted driving. Often, software engineers just take results from a code copilot for granted. Yet, each single result from such GenAI needs manual in-depth verification. Copiloting is a critical AI competence that can only evolve over time with continuous training and feedback.

Figure 3 displays an AI competence grid, which splits between explicit and implicit competences. Two major new competences are in high demand, namely meta knowledge on how to utilize and assess AI and judgement, i.e., capability to review and judge AI results. In working with AI departments and companies, we typically first evaluate the priorities within each quadrant to avoid having one solution that fits no need. We then assess people, distinguishing roles and managerial levels. Self-rating is possible, typically done by using case studies. For instance, a

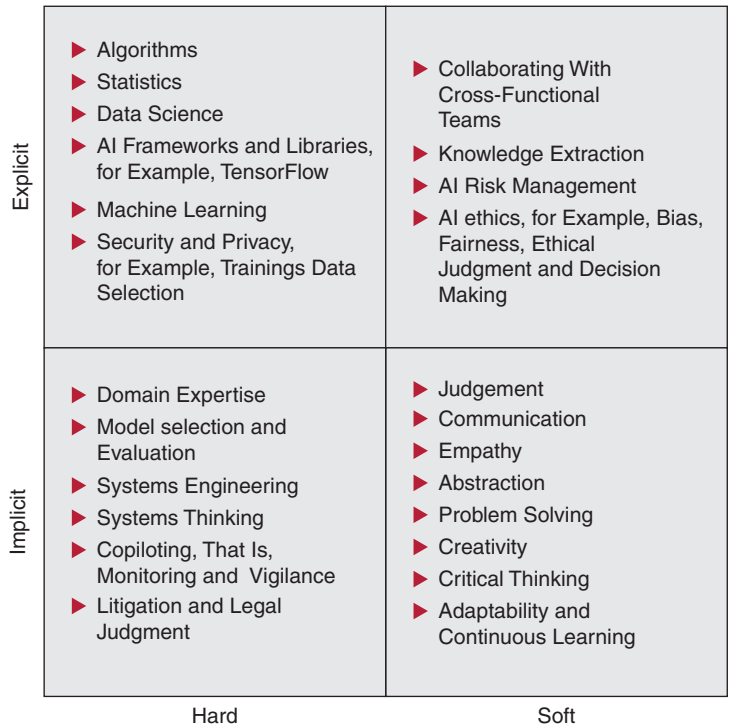

FIGURE 3. The AI competence grid.

development engineer might be strong in explicit hard skills but might need support in the other three domains. On the other hand, a team lead might be strong in the upper half, that is, hard and soft explicit competences, but have never really worked on the lower half. With such a stepwise approach, you can tailor a competence program to specific needs.

To understand industry competence challenges, we compare the AI competences in Figure 3 with their availability today:

- *Skills that are used today and will further grow*: hard competences, such as data science, algorithms, security, and AI frameworks, and soft competences, such as problem solving, abstraction, and communication
- *Skills that are scarce today among professionals and will be in strong future demand*: systems engineering, model selection and evaluation, copiloting, and most soft competences, such as empathy and ethics.

Winning the AI competence race will remain a challenge for companies across industries. The challenges are even greater with the second category of skills that are weak today and will be in high future demand.

Special attention in the industry context should be given to grow AI ethical competences. AI is like a mirror of the values and thinking of its human creators, as there are no inherent "machine values." The morality of AI is based on the data and algorithms developed and provided by humans. Engineers prefer the simple golden rule. The introduction of "robot rules" can therefore create new conflicts, as these rules are based on human concepts and ethical ideas, which are often interpreted differently and can vary. Isaac Asimov was among the first to try, yet could only prove that moral rules fail in the face of complex technology and malevolence. His first robot law already shows this inherent weakness. Rules such as "a robot must not harm a human" are insufficient, as there is no universal agreement about the meaning of "harm." How should an AI deal with conflict situations involving harm to different people, as every action can cause harm if thought through to the end—even doing nothing. Obviously, ethics is a complex and multilayered concept that manifests itself in the context of cultural, social, and individual norms. It is difficult to transfer this to IT systems, as ethics is not about simple rules but requires a compass to embed moral decisions in a situational and cultural context.

GUIDE TO GROW

TAKEAWAYS ON GROWING AI CORPORATE COMPETENCES:

- Provide a structure for individual competence growth that balances persistence in achieving targets with flexibility to accommodate urgent project and technology needs.
- Facilitate collaborative team learning, such as mentoring and apprenticing, to address learning goals beyond obvious explicit hard skills.
- Design online learning formats to connect AI skills to everyday work.
- Use company-specific competence assessment and rating schemes to allow corporate visibility of competence goals, achievements, and success factors.

TAKEAWAYS FOR GROWING YOUR INDIVIDUAL COMPETENCES:

- Enroll in AI courses from relevant learning platforms. Give preference to online or distant "classroom" formats to avoid a bias on your own current situation.
- Engage in practical projects. Join voluntary projects in your own environment or start a specific extra step.
- Practice hackathons and coding and algorithmic challenges on online platforms, such as LeetCode, CodeWars, CodeSignal, and HackerRank.
- Join AI communities and attend conferences, such as the IEEE Conference on Artificial Intelligence, to grow your network and stay updated on industry trends.
- Invest in your own AI ethics competences to understand the implications of AI and ensure responsible use. Use professional societies, such as IEEE, as well as ethics societies, such as the World Ethic Ambassador program.

TRANSFER QUESTIONS:

- How can you grow your business model by using specific AI techniques?
- Which ethical considerations should you consider within your AI systems? Think about training data, bias, and fake results.
- Does a digital application promote humanity, or is it an obstacle? Create examples for both perspectives and synthesize.
- What risks do you see within your current projects? Think about your intellectual property, transparency, and corner cases when testing.
- Which standards do you need to deploy? Consider legislative demands and individual liability.
- Are you prepared to take personal responsibility, such as whistleblowing or escalating, if your project, client, or company violates ethics guidelines?

AI FOR LEARNING: A CASE STUDY

Individual and corporate-level competence growth can be facilitated and improved with GenAI based e-learning (see "Guide to Grow"). GenAI can transform training from generic offerings into a personalized interactive

experience that addresses specific skill gaps and prepares employees for upcoming challenges. We have been using AI-based training for both gap analysis and providing tailored learning modules. Here are some recommendations from our trainings:

- *Step 1—Initial gap analysis*: Develop a tailored curriculum that identifies specific learning needs and skill gaps versus a defined role and competence scheme. This is done with standardized questions and small case studies.
- *Step 2—Training creation*: Synthesize and aggregate with GenAI existing training materials, corporate rules, standards, and so on into individualized training formats that coherently address the observed gaps from step 1.
- *Step 3—Training delivery*: Enhance existing e-learning platforms with GenAI to package the contents of step 2. Have GenAI provide digestible training content that is adjusted automatically to available times, gaps, and preselected learning goals. Map the generic training contents of step 2 to current project needs to improve the user experience. Adapt to individual learning styles, such as questions first, then content, presentations, coherent text, and so on.
- *Step 4—Feedback*: Utilize GenAI to offer immediate data-driven feedback on learning progress. Facilitate feedback-driven automatic updates on teaching materials, such as restating learning questions; providing answers to simple questions, which heavily depends on your large language model (LLM) and underlying hardware platform; and selecting sources to dig deeper into specific content.

One warning for industry users. Do not try to do such competence programs with public cloud-based LLMs. They will obscure sources, release your own intellectual property, hallucinate on content, and oversimplify results. Rather, use an on-premises local tool that is tailored to specific contents needed by roles, such as product manager, tester, or systems engineer.[6]

WHERE DO WE GO FROM HERE?

AI is changing society and industry more than any other innovation since the invention of computers. It improves productivity in various ways, for example, by automating repetitive tasks; increasing safety in critical activities, such as in medicine or decontamination; and facilitating fatigue-free constant efficiency, such as in self-driving vehicles. At the same time, AI creates completely new risks, as it is neither transparent, deterministic, nor explainable. Fake malicious information and false images, the arbitrary collection of inadequate training data, and automatic AI tools for sabotage and cybercrime are just a few examples that show that AI needs critical competences that we rarely possess today.

AI needs new hard and soft competences, both explicit and implicit. Engineers with the right AI competence mix, hard and (!) soft, are in demand across domains, from IT to the IoT, from app development to big systems, from automotive to robotics. There is even a legal obligation to grow professional competences because each major standard demands learning and proof of learning.

Growing AI competence is not just an individual exercise but also a need for companies and societies. When famous first driver Bertha Benz was driving her car, there were no regulations. They were not necessary for many years because those who had a car were highly competent and intelligent people. With more people driving, regulations were implemented to the point where it was required by law to have a driver's license to prove maturity.

Insufficient AI competences further widen the digital divide along a variety of lines, such as knowledge, risk awareness, critical thinking, culture, and, not least, our societies at large. Jeff Horwitz, in his assessment of AI companies, concludes that Facebook senior management and its chief executive officer have chosen to prioritize growth and engagement over any other objective.[9] Ethics are for Sunday prayers, while malicious business models have become the daily routine.

Many companies consider training and competence growth a cost factor rather than an investment. During economic downturns, trainings are canceled first. During the pandemic lockdown, many clients canceled trainings to focus on the product pipeline. In turn, engineers were demotivated and lacked the necessary energy and know-how to master the crisis. A vicious circle had started. Insufficient competence focus in companies is a recipe for failure. Lower-quality and less innovative products inevitably increase the red ocean, which further fuels the pressure and thus does not allow investments in competence growth.

> AI needs critical competences that go way beyond typical engineering and computer science.

Mastering AI competences enables professionals to develop robust, effective, and ethical AI systems that deliver value. Continuous learning of hard and soft AI competences and adaptability to fast technology changes are crucial in this ever-evolving field, ensuring that practitioners, scientists, and industry managers remain proficient and relevant. Without critical AI expertise, we cannot assess the impact on society and ethical dilemmas at all. Famous philosopher Immanuel Kant, whose

300th birthday we currently commemorate, wisely observed, "We need to learn how to think, and not to learn thoughts." He underlined what distinguishes human intelligence from AI, namely, that we learn to grow and not just statistical patterns.

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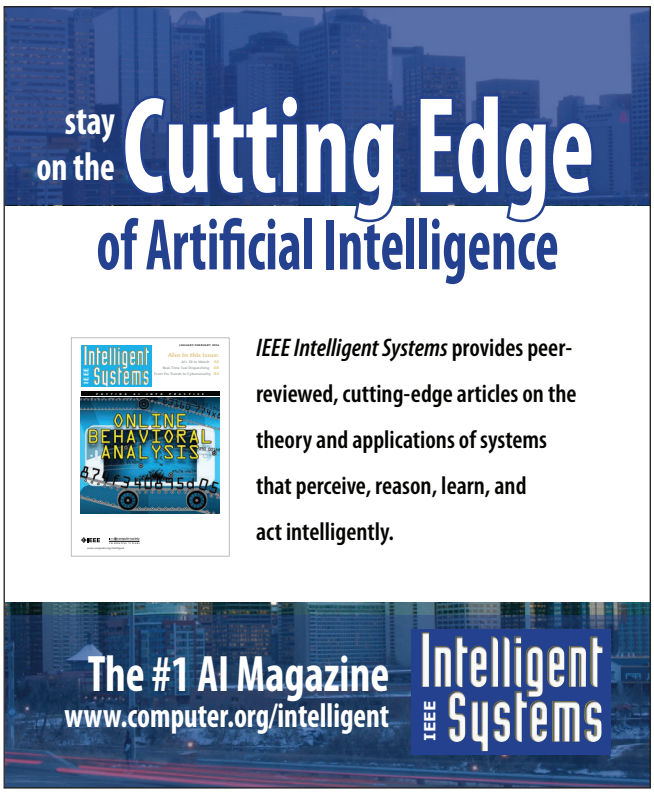

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